

CSE 340: Computer Architecture

[Spring 2025]

Quiz - 3 :: Set - A

Time: 15 min

Marks: 15

Name: _____

ID: _____ Section: _____

Given the following values encoded in IEEE-754 single-precision format:

- Initial velocity, $u = 0x40B00000$
- Time, $t = 0x40C66666$
- Acceleration, $a = 0x40533333$

Find the value of displacement.

Constraints:

1. All arithmetic operations must be performed using binary floating-point arithmetic.
2. The final answer must be encoded in IEEE-754 double-precision format and provided in hexadecimal representation.

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Quiz - 3 :: Set - B

Time: 15 min

Marks: 15

Name: _____

ID: _____ Section: _____

Given the following values encoded in IEEE-754 single-precision format:

- Initial velocity, $u = 0x41080000$
- Time, $t = 0x40A00000$
- Acceleration, $a = 0xC04CCCCD$

Find the value of displacement.

Constraints:

3. All arithmetic operations must be performed using binary floating-point arithmetic.
4. The final answer must be encoded in IEEE-754 double-precision format and provided in hexadecimal representation.